

Lecture 7 Exercises

Exercise 1

- Construct an ER diagram for a car-insurance company. The model includes
 - each customer is described by a unique id, a name, and an address;
 - each car is described by a unique vehicle identification number, the manufacturer, and the year of production;
 - each accident is described by a unique report number, the date and the location when the accident happens;
 - each customer owns one or more cars;
 - every car is owned by at exactly one customer;
 - each accident is only responsible to the owner of the car, no matter who drives;
 - it is possible that multiple cars involve in a same accident and the amount of insurance claim for different cars are different.
- List all assumptions that you make.
- And convert your ER diagram to a logical design.

Exercise 2

- Construct an ER diagram for a hospital with a set of patients and a set of medical doctors, which models
 - each patient has a unique id, a name, and an insurance;
 - each doctor has a unique id, a name, and the specialization;
 - each test has a unique id, a name, the date, the time, and the result of the test;
 - every patient is cared by one or multiple doctor;
 - every test is performed by one or multiple doctor;
 - every test is done for exactly one patient.
- List all assumptions that you make.
- And convert your ER diagram to a logical design.

Exercise 3

- Create an ER diagram for a recording company with the following requirements:
 - Each musician has an unique SSN, a name, an address, and a phone number.
 - Each instrument has a name (e.g., guitar) and a musical key (e.g., C).
 - Each album has a title, a copyright date, a format (e.g., CD), and an album identifier.
 - Each song has a title and an author.
 - Musicians may play many instruments, and an instrument may be played by many musicians.
 - Each album has a number of songs on it, but no song may appear on more than one album.
 - Each song is performed by one or more musicians, and a musician may perform a number of songs.
 - Each album has exactly one musician who acts as its producer. A musician can of course produce several albums.

End of Lecture 7